

Youssef Hefny

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EDUCATION

Egyptian Japanese University of Science and Technology (E-JUST)

Bachelor of Science in Computer Science — Specialization in Artificial Intelligence

Alexandria, Egypt

Expected June 2026

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Software Engineering, Parallel Computing, Data Structures & Algorithms

EXPERIENCE

National Institute of Telecommunications (NTI)

Data Analysis Intern

Alexandria, Egypt

July 2024 – October 2024

- Engineered end-to-end data pipelines in Python and R using Power Query for data transformation, cleaning, and feature extraction across multiple telecom datasets.
- Developed interactive KPI dashboards in Power BI and Tableau, enabling stakeholder visibility into operational metrics and reducing manual reporting overhead.
- Applied statistical analysis techniques in Python and R to surface data-driven insights; presented findings to supervisors via structured Excel reports.
- Collaborated in a professional analyst environment, translating business requirements into analytical deliverables under real-world project deadlines.

PROJECTS

MCP Client & Server | Python, MCP SDK, Anthropic API

March 2026

- Designed and implemented a full Model Context Protocol (MCP) client-server architecture, exposing custom tools to an LLM agent over a structured communication layer.
- Integrated the Anthropic API to power tool-augmented conversational flows; demonstrated agentic AI patterns including tool routing, multi-step reasoning, and context management.

RAG Chatbot with Web Scraping | Python, FAISS, LangChain, BeautifulSoup

March 2026

- Built a retrieval-augmented generation (RAG) chatbot that ingests knowledge exclusively via web scraping, eliminating the need for manual dataset preparation.
- Engineered dual operating modes (Extraction and Generation) plus an Auto mode, with a per-answer confidence scoring system to surface answer reliability.
- Added on-demand URL scraping feature allowing users to inject live web content into the chatbot knowledge base at runtime.

Parallel Computing Web Application | Python, Flask, MPI

May 2025

- Built a Flask web application benchmarking serial vs. parallel task execution via MPI, with a real-time comparison UI demonstrating measurable speedup factors.
- Produced full software engineering documentation including UML diagrams, system architecture, and requirements specification as part of a formal academic deliverable.

Deepfake Detection System | Python, PyTorch, wav2vec 2.0, CLIP ViT, SVD

2025

- Developed a multimodal deepfake detection pipeline combining video analysis (SVD + CLIP ViT feature extraction) and audio analysis (wav2vec 2.0 with CNN-based spectrogram classification).
- Deployed as a full-stack web application (FastAPI + Next.js) with end-to-end inference pipeline; resolved cross-platform compatibility and environment configuration issues.

Human Activity Recognition (HAR) | Python, TensorFlow, LSTM, GRU 2024

- Built sequential deep learning models (LSTM and GRU) to classify human physical activities from sensor time-series data, achieving robust multi-class recognition.
- Designed preprocessing pipeline for windowing, normalization, and sequence batching of raw accelerometer/gyroscope data.

Weather Prediction — Time-Series Forecasting | Python, TensorFlow, LSTM, GRU 2024

- Engineered LSTM and GRU networks for multi-step weather forecasting; compared architectures on sequence length, loss, and generalization performance.

SKILLS & INTERESTS

- **Programming:** Python, C, C++, R, SQL
- **AI / ML:** Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP)
- **Frameworks & Libraries:** TensorFlow, PyTorch, HuggingFace Transformers, LangChain, spaCy, FAISS, Flask, FastAPI, BeautifulSoup, OpenCV
- **DevOps & Tools:** Docker, Kubernetes, Git, GitHub, Jupyter, Google Colab
- **Data & BI:** Power BI, Tableau, Excel, Power Query
- **Certifications:** Datacamp Containerization and Virtualization with Docker and Kubernetes (2025), Anthropic Introduction to MCP (2026), Anthropic MCP: Advanced Topics (2026)
- **Languages:** Arabic (Native), English (Fluent)